AMRITA VISHWA VIDYAPEETHAM

23MNG336 : Operations Research

Multi-Objective Stock Selection & Portfolio Construction

Team B11
Gowtham S D - CB.AI.U4AID23113
Guhan K B - CB.AI.U4AID23114
Vaishnavi Gupta - CB.AI.U4AID23149
Vyshnav Kumar S - CB.AI.U4AID23151

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Introduction

- Stock Selection Challenge: The large Indian stock universe makes manual selection difficult, since, higher returns may increase risk and reduce diversification.
- Optimization Limitations: Traditional methods focus mainly on risk and return, overlooking multiple portfolio objectives.
- Proposed Approach: Combines Machine Learning and Operations Research for systematic, diversified portfolio construction.

Literature review

S NO.	TITLE AND YEAR	KEY CONTRIBUTION	LIMITATIONS
1	Multi-Objective Portfolio Optimization via NSGA-III (Thomas & Douglas, 2025)	Compares NSGA-III against Mean-Variance across risk, return, skewness and kurtosis.	Evaluated on global assets, not the Indian market; no liquidity or turnover objective.
2	Optimizing Portfolios with Modified NSGA-II Solutions (Lou, 2023)	Refines NSGA-II with dynamic mutation and better initialization; uses MCMC for long-horizon forecasts.	Backtested returns remain unstable; sensitive to the chosen risk measure.
3	EvoFolio: Portfolio Optimization via Multi-Objective Evolutionary Algorithms (2024)	NSGA-II framework that jointly maximizes yield and minimizes risk on real stock datasets.	Short (3-year) evaluation window; only two objectives considered.
4	Portfolio Optimization Using NSGA-II — Tehran Exchange (Ramezani et al., 2024)	Combines Markowitz semi-variance with NSGA-II under real-world constraints on 132 stocks.	Single-market case study; constraints are not tuned for NSE-scale universes (1,700+ stocks).

Research Gap

- Limited Objectives: Most studies focus mainly on risk and return.
- Small-Scale Testing: NSGA-II/III are often tested on small datasets.
- ML Gap: ML methods are rarely integrated with portfolio optimization.
- Reproducibility Gap: Few end-to-end, open NSE pipelines exist.
- Indian-Market Gap: Limited validation across long-term NSE data.

Objective

- Develop a systematic framework for stock selection and portfolio construction.
- Calculate key financial metrics and apply K-Means clustering for stock grouping.
- Use Integer and Goal Programming to select and allocate stocks while balancing risk, return, and diversification.
- Evaluate the portfolio through out-of-sample backtesting against baseline strategies.

Dataset Details

India Stock Data – NSE

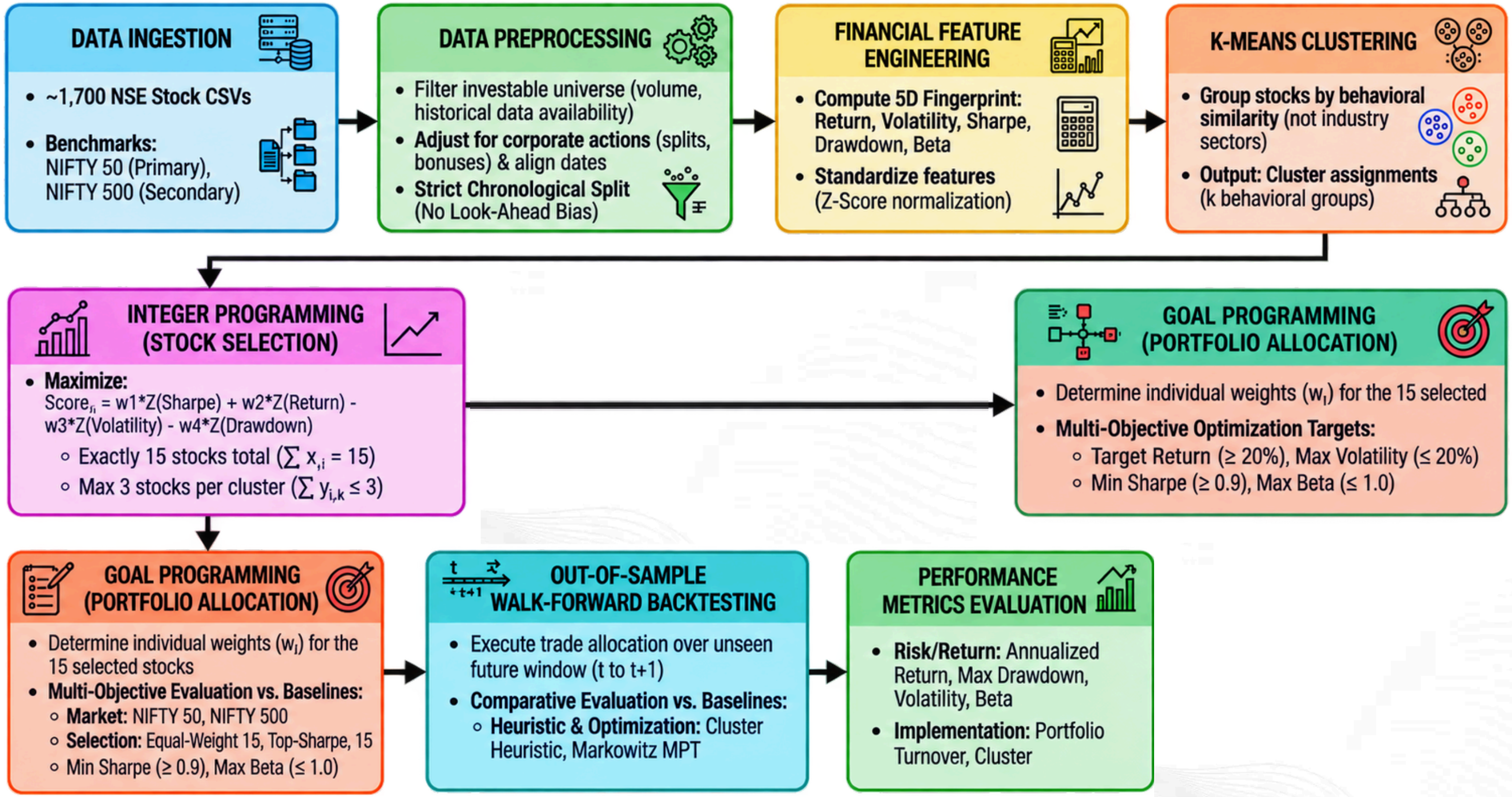
- **Source**: Kaggle — NSE India Stock Data
- **Coverage:** January 1990 – June 2021

Dataset Components

- **NSE Equity Data**: Daily OHLCV (Open, High, Low, Close, Volume) data covering 1,700+ stocks, forming the core market dataset.
- **NSE Index Data**: 50+ market, sectoral, and thematic indices used for benchmark comparison, beta calculation, and market-regime analysis.
- **NSE Symbols**: Ticker-to-company mapping used to clean and define the investable stock universe.

Role in the Project: Provides the foundation for stock screening → feature engineering → clustering → multi-objective portfolio optimization → backtesting.

Methodology - Flow



Methodology : Data & Financial Feature Engineering

Financial Data

- Historical daily NSE stock data: OHLC, Volume, VWAP
- NIFTY 50 used for market-relative calculations and primary benchmarking.

Financial Features

- Return — Historical performance
- Volatility — Return variability
- Sharpe Ratio — Risk-adjusted performance
- Beta — Sensitivity to NIFTY 50
- Maximum Drawdown — Downside risk
- Liquidity — How easily the stock can be traded.

Methodology : K-Means Clustering

- Stocks are grouped according to financial behavior, not industry.
- Clustering features:
  - Return
  - Volatility
  - Sharpe Ratio
  - Beta
  - Maximum Drawdown
  - Liquidity
- Similar stocks are assigned to the same behavioral cluster.
- Cluster membership is later used as a diversification constraint.
- This prevents the optimizer from selecting 15 highly similar stocks.

K-Means is used for diversification, not price prediction.

Methodology : Integer Programming

- Binary variable:
  - $x_i = 1 \rightarrow$ selected
  - $x_i = 0 \rightarrow$ not selected

Diversification Constraints:

- Exactly 15 stocks
  - Maximum 3 stocks per cluster
  - At least 5 clusters represented
- **Output:** The optimal set of 15 stocks.

Selection score formula:

$$\mathbf{Score}_i = w_1 Z(\mathbf{Sharpe}_i) + w_2 Z(\mathbf{Return}_i) + w_5 Z(\mathbf{Liquidity}_i) - w_3 Z(\mathbf{Volatility}_i) - w_4 Z(\mathbf{Drawdown}_i)$$

$$w_1, w_2, w_3, w_4, w_5 \geq 0, \quad w_1 + w_2 + w_3 + w_4 + w_5 = 1$$

Methodology : Goal Programming

- Determines the weight of each selected stock.
- Balances multiple portfolio-level objectives:
  - Target Return
  - Maximum Volatility
  - Minimum Sharpe Ratio
  - Maximum Beta
- Portfolio risk considers covariance between stocks.

Integer Programming → Which stocks?

Goal Programming → How much in each stock?

Methodology : Backtesting & Comparison

Strategies:

- Proposed: K-Means + IP + GP
- Equal-Weight Same 15
- Top-Sharpe 15
- Top-Return 15
- Random 15

Benchmarks:

- NIFTY 50
- NIFTY 500

Evaluation:

- Cumulative Return
- Volatility
- Sharpe & Sortino Ratio
- Maximum Drawdown
- Beta
- Turnover

Primary focus: Consistent risk-adjusted out-of-sample performance rather than return alone.

Methodology : Feedback Mechanism

Walk-Forward Backtesting:

- Use a 4-year rolling window from 1990 onward.
- Apply year-wise/recency weighting within each window.
- Recalculate features, re-run K-Means, and re-optimize the portfolio.
- Test on the following 1-year out-of-sample period.
- Roll the window forward and repeat until the latest period.

Important:

- Each window is independently re-estimated.
- No future information is used in portfolio construction.

Tools & Techniques

Component	Key Tasks	Tools
Data Engineering	Cleaning, alignment, corporate actions	Python, Pandas, NumPy
Feature Engineering	Return, Volatility, Sharpe, Beta, Drawdown	Pandas, NumPy
Standardization & Clustering	Z-score + behavioral K-Means	Scikit-learn
Integer Programming	Select 15 stocks; cluster constraints	Pyomo / PuLP, Gurobi
Goal Programming	Optimize portfolio weights	Pyomo / CVXPY
Walk-Forward Backtesting	4-year rolling window + 1-year test	Pandas, NumPy
Evaluation & Baselines	Risk-adjusted metrics + Markowitz	PyPortfolioOpt, CVXPY, Matplotlib

References

[1] H. Markowitz, “Portfolio Selection,” Journal of Finance, 1952.

[2] Thomas and Douglas, “Multi-Objective Portfolio Optimization: An Application of NSGA-III,” Risks (MDPI), 2025.

[3] K. Lou, “Optimizing Portfolios with Modified NSGA-II Solutions,” IEEE ICDSCA, 2023.

[4] “EvoFolio: A Portfolio Optimization Method Based on Multi-Objective Evolutionary Algorithms,” Neural Computing and Applications, 2024.

[5] “An Integrated Framework for Classification and Selection of Stocks for Portfolio Construction: Evidence from NSE, India,” Decision Making: Applications in Management and Engineering.

[6] Kaggle Dataset: “NSE India Stock Data (1990–2021),” stoicstatic, kaggle.com/datasets/stoicstatic/india-stock-data-nse-1990-2020.